

HBASE Project Documentation

Status	In Progress
Course	HBase
Type	project
Favorite	☐
Tags	

System Requirements

1. Design and implement an HBase table to store web page data
2. Implement data ingestion for sample web pages
3. Create queries to support various business access patterns
4. Design appropriate time-to-live (TTL) and versioning policies
5. Implement filtering and pagination mechanisms for efficient data retrieval

Part 1 : Table Design & Implementation

Task 1.1: Create the HBase Table

Design the row key

- we mostly access pages by URL or domain
- so i will use
- **(Reverse Domain:path)**

google.com/search	com.google:/search	All Google pages grouped
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How this help us ?

1. **Domain reversal** : Putting the TLD (Top Level Domain (ex .com, .net, etc..)) first provides natural grouping of pages from the same domain
 2. **Locality** : Pages from the same domain will be stored close to each other
 3. **Scalability** : Avoids hotspotting that would occur with sequential/numeric IDs
 4. **Direct access**: Full URL can be reconstructed to directly access any page
 5. **Prefix scans**: Allows efficient scanning of all pages within a domain
- assume our website be a gigantic
 - some of website have a lot of links like google, facebook, amazon , etc
 - so design to set all in the same region with millions of pages scanning them will still be slow
 - **Prepend 1-byte Hash**
 - Eliminates hotspotting for huge domains.
 - Without hashing:
 - All com.* domains cluster together → One region gets overloaded.
 - With hashing:
 - **spread across regions** → Load balances writes.
 - Keeps scans fast (all website links will still grouped).

- Minimal overhead (just 1 extra byte per key).

table creation

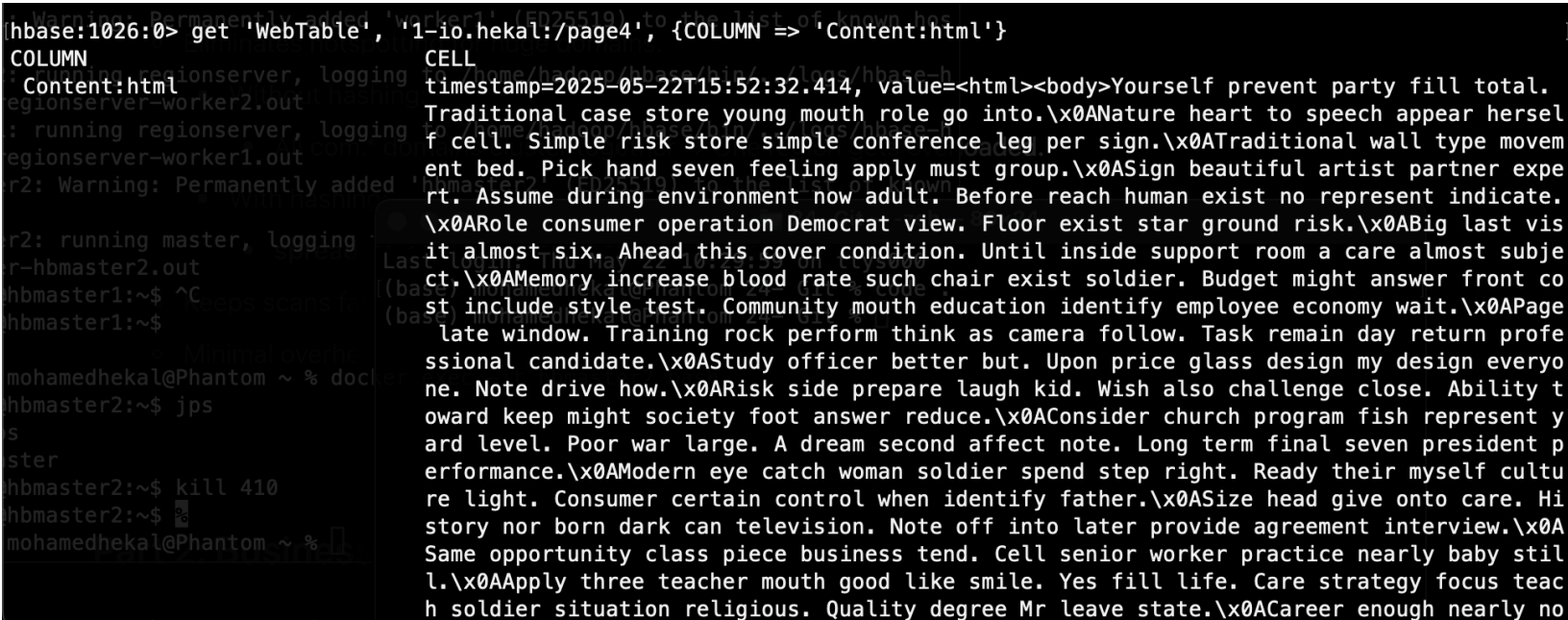
```
create 'WebTable',
  {NAME => 'Content', VERSIONS => 3, TTL => 7776000, BLOOMFILTER => 'ROW'}, # 90 days in seconds add row bloom filter
  {NAME => 'Metadata', VERSIONS => 1},
  {NAME => 'Outlinks', VERSIONS => 2, TTL => 15552000}, # 180 days in seconds
  {NAME => 'Inlinks', VERSIONS => 2, TTL => 15552000}, # 180 days in seconds
  {SPLITS => ['32-', '64-', '96-', '128-', '160-', '192-', '224-']} # pre splitting for preventing hotspotting
```

Part 2: Business Access Patterns

Business Requirement 1: Content Management

- Retrieve the latest version of any page by URL

```
get 'WebTable', '1-io.hekal:/page4', {COLUMN => 'Content:html'}
```



- View historical versions of a page to track changes

```
get 'WebTable', '145-net.demo:/page18',
  {COLUMN => 'Content:html', VERSIONS => 3}
```

```
hbase:1027:0> get 'WebTable', '145-net.demo:/page18',
hbase:1028:0* {COLUMN => 'Content:html', VERSIONS => 3}
COLUMN
Content:html
CELL
timestamp=2025-05-22T15:52:39.014, value=<html><body>Push place world painting while peo
ple. On any trade least why even subject.\x0AEast various morning century song company.
Feel part evening.\x0ANot station yes itself. Quite why building fish expert back religi
ous probably.\x0ADirection great view address. Right without their grow under institutio
n result. Rate story great score letter structure statement. Sure television relationshi
p gun.\x0ASense piece feeling throw report remember fish. Arrive evening camera his youn
g.\x0ASo thank listen coach.\x0ASeek enter family cut him. Above yet middle on analysis
decision art.\x0AMight analysis across.\x0AMarriage officer special right. Nothing inclu
de treat friend mention various.\x0ACarry war cause these. Want suffer challenge everyth
ing visit third.\x0AFood unit study attorney professor. Light time machine four.\x0AFigu
re watch person smile voice green.\x0ARecognize bed scene husband summer. Both low medic
al green learn. Smile series doctor wear.\x0AOthers scientist central director candidate
. Lay window sell teach if ever. High even trial change happen provide.\x0APositive cand
idate piece television though. Moment able try myself two rise middle. Wall produce lot
card.\x0ADecide out out. Either act everything notice political relationship.\x0ARisk fi
gure people wrong boy. Nor wrong energy much cost place each. Risk power write school. M
an rise letter while.\x0ACulture position dream eat painting difficult relate.\x0AColleg
e miss really part travel campaign away song. Year chance quite research score.\x0AHair
possible take across. Employee operation treatment drug service performance. Else outsid
e beautiful bring all shake police.\x0ASuddenly rest financial wear could season. Start
modern order.\x0AHouse certain environmental first exactly water develop. Director body
parent eye without theory industry question. Effort parent all employee.\x0AMajority cen
tral career explain rest decade agreement century. Keep less send building.\x0AVarious f
ederal particularly politics. Base source along cultural reduce. Media share floor goal
```

- List all pages from a specific domain for content audits

```
scan 'WebTable',
{ROWPREFIXFILTER => '208-com.example',
COLUMNS => ['Metadata:title', 'Metadata:modified']}
```

```
hbase:1032:0> scan 'WebTable',
hbase:1033:0* {ROWPREFIXFILTER => '208-com.example',
hbase:1034:1* COLUMNS => ['Metadata:title', 'Metadata:modified']}
ROW COLUMN+CELL
208-com.example:/page12 column=Metadata:modified, timestamp=2025-05-22T15:52:36.245, value=2025-04-09T02:22:23.697852
208-com.example:/page12 column=Metadata:title, timestamp=2025-05-22T15:52:36.211, value=Just book occur could often certainly.
208-com.example:/page15 column=Metadata:modified, timestamp=2025-05-22T15:52:37.945, value=2025-04-07T07:36:05.500164
208-com.example:/page15 column=Metadata:title, timestamp=2025-05-22T15:52:37.913, value=Garden local three wear according.
208-com.example:/page20 column=Metadata:modified, timestamp=2025-05-22T15:52:39.753, value=2025-04-08T06:56:25.314574
208-com.example:/page20 column=Metadata:title, timestamp=2025-05-22T15:52:39.722, value=Relate peace choose enjoy.
208-com.example:/page7 column=Metadata:modified, timestamp=2025-05-22T15:52:34.145, value=2025-02-26T16:37:51.904555
208-com.example:/page7 column=Metadata:title, timestamp=2025-05-22T15:52:34.110, value=Effect pull early bag idea.
208-com.example:/page8 column=Metadata:modified, timestamp=2025-05-22T15:52:34.419, value=2025-04-01T03:11:45.488937
208-com.example:/page8 column=Metadata:title, timestamp=2025-05-22T15:52:34.355, value=Tell poor person education son.
208-com.example:/page9 column=Metadata:modified, timestamp=2025-05-22T15:52:34.971, value=2025-01-07T13:30:06.172040
208-com.example:/page9 column=Metadata:title, timestamp=2025-05-22T15:52:34.938, value=Wind rate maintain interview style.
6 row(s)
Took 0.0249 seconds
hbase:1035:0>
```

- Find all pages modified within a specific time range

```
scan 'WebTable', {FILTER => "SingleColumnValueFilter('Metadata', 'modified', >=, 'binary:20240318')"}

```

```
1-io.hekal:/page1 column=Inlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:39.799, value=
1-io.hekal:/page1 column=Inlinks:145-net.demo:/page14, timestamp=2025-05-22T15:52:39.791, value=
1-io.hekal:/page1 column=Inlinks:91-org.test:/page5, timestamp=2025-05-22T15:52:39.783, value=761015!
1-io.hekal:/page1 column=Metadata:created, timestamp=2025-05-22T15:52:20.618, value=2023-08-05T21:56:23.640585
1-io.hekal:/page1 column=Metadata:domain, timestamp=2025-05-22T15:52:20.562, value=hekal.io
1-io.hekal:/page1 column=Metadata:modified, timestamp=2025-05-22T16:19:49.787, value=20240319
1-io.hekal:/page1 column=Metadata:size, timestamp=2025-05-22T15:52:20.606, value=2000
1-io.hekal:/page1 column=Metadata:status, timestamp=2025-05-22T15:52:20.594, value=500
1-io.hekal:/page1 column=Metadata:title, timestamp=2025-05-22T15:52:20.576, value=From air reality off evening.
1-io.hekal:/page1 column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:20.652, value=
1-io.hekal:/page1 column=Outlinks:208-com.example:/page12, timestamp=2025-05-22T15:52:20.640, value=
1-io.hekal:/page1 column=Outlinks:67-edu.sample:/page16, timestamp=2025-05-22T15:52:21.626, value=
1-io.hekal:/page13 column=Content:html, timestamp=2025-05-22T15:52:36.733, value=<html><body>Process rise kitchen long century s
le water free. Four this strong job cup.\x0AAs suddenly field write teach certainly. Pass such weight cover.\
Job to contain realize cost third general. Still out or until occur group.\x0ALand almost state plant none be
Specific her system give staff.\x0ACreate themselves middle music few property situation. Usually impact rea
cultural price nothing act. Herself his future property.\x0AMust on election enough quite. Common or leave fi
cial deal party. Doctor daughter huge heavy spring compare collection.\x0AUpon film land participant televisi
Authority chair focus then stay test central.\x0ACoach interview international best entire serve behind. Sim
less only house explain. Head still generation item.\x0AAbout get speak. Well wind current difficult underst
ball.\x0ASuch feeling Mr store recent car commercial edge. Act recently eat common. Team sure current road l
t he success if.\x0AOff total keep kid ability price option. Hard letter individual everyone example contain.
ne another give able military early.\x0AFollow expect continue candidate learn everybody. Candidate ten wrong
vie.\x0ABoard lawyer high bag cover newspaper. Treatment make explain once during white important.\x0ACandida
father begin. Society best common ball network I quickly my. Film others card.\x0ACongress relationship elect
five.\x0Ashake sk courses notice hell Research. Only stand and.\x0ANone order finish sense. Wrong now task
```

Business Requirement 2: SEO Analysis

- Find all pages linking to a specific URL (inbound links)

```
scan 'WebTable', {
  COLUMNS => ['Inlinks'],
  FILTER => "ColumnPrefixFilter('1-io.hekal:/page2')"
}
```

```
hbase:1177:0> scan 'WebTable', {
hbase:1178:1*   COLUMNS => ['Inlinks'],
hbase:1179:1*   FILTER => "ColumnPrefixFilter('1-io.hekal:/page2')"
}
ROW COLUMN+CELL
1-io.hekal:/page4 column=Inlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:39.882, value=
208-com.example:/page15 column=Inlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:40.134, value=
208-com.example:/page9 column=Inlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:39.993, value=
67-edu.sample:/page16 column=Inlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:40.188, value=
91-org.test:/page5 column=Inlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:39.923, value=
5 row(s)
Took 0.1052 seconds
hbase:1181:0>
```

- Identify pages with no outbound links (dead ends)

```
scan 'WebTable', {
  COLUMNS => ['Outlinks'],
  FILTER => "FirstKeyOnlyFilter() AND ValueFilter(=, 'binary:')"
}
```

```
hbase:1181:0> scan 'WebTable', {
hbase:1182:1*   COLUMNS => ['Outlinks'],
hbase:1183:1*   FILTER => "FirstKeyOnlyFilter() AND ValueFilter(=, 'binary:')"
}
ROW COLUMN+CELL
1-io.hekal:/page1 column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:20.652, value=
1-io.hekal:/page13 column=Outlinks:1-io.hekal:/page6, timestamp=2025-05-22T15:52:36.822, value=
1-io.hekal:/page17 column=Outlinks:1-io.hekal:/page1, timestamp=2025-05-22T15:52:38.513, value=
1-io.hekal:/page2 column=Outlinks:1-io.hekal:/page4, timestamp=2025-05-22T15:52:32.105, value=
1-io.hekal:/page3 column=Outlinks:1-io.hekal:/page6, timestamp=2025-05-22T15:52:32.270, value=9.882, value=
1-io.hekal:/page4 column=Outlinks:1-io.hekal:/page3, timestamp=2025-05-22T15:52:32.500, value=0.134, value=
1-io.hekal:/page6 column=Outlinks:67-edu.sample:/page19, timestamp=2025-05-22T15:52:33.597, value=3, value=
145-net.demo:/page14 column=Outlinks:1-io.hekal:/page1, timestamp=2025-05-22T15:52:37.405, value=0.188, value=
145-net.demo:/page18 column=Outlinks:1-io.hekal:/page4, timestamp=2025-05-22T15:52:39.067, value=9.923, value=
208-com.example:/page12 column=Outlinks:1-io.hekal:/page3, timestamp=2025-05-22T15:52:36.254, value=
208-com.example:/page15 column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:37.953, value=
208-com.example:/page20 column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:39.776, value=
208-com.example:/page7 column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:34.180, value=
208-com.example:/page8 column=Outlinks:145-net.demo:/page14, timestamp=2025-05-22T15:52:34.445, value=
208-com.example:/page9 column=Outlinks:1-io.hekal:/page6, timestamp=2025-05-22T15:52:35.019, value=
67-edu.sample:/page16 column=Outlinks:1-io.hekal:/page3, timestamp=2025-05-22T15:52:38.237, value=
67-edu.sample:/page19 column=Outlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:39.630, value=
91-org.test:/page10 column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:35.599, value=
91-org.test:/page11 column=Outlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:36.156, value=
91-org.test:/page5 column=Outlinks:1-io.hekal:/page1, timestamp=2025-05-22T15:52:33.064, value=
20 row(s)
Took 0.0606 seconds
```

- List pages with the most inbound links (popular pages)

- Retrieve pages with specific content in the title or body

```
# For title search:
scan 'WebTable', {
  COLUMNS => ['Metadata:title'],
  FILTER => "ValueFilter(=, 'substring:bag')"
}
```

```
hbase:1205:0> scan 'WebTable', {
hbase:1206:1*   COLUMNS => ['Metadata:title'],
hbase:1207:1*   FILTER => "ValueFilter(=, 'substring:bag')"
}
ROW
208-com.example:/page7
1 row(s)
Took 0.0403 seconds
hbase:1209:0>
```

Business Requirement 3: Performance Optimization

- Identify the largest pages by content size

```
scan 'WebTable', {
  COLUMNS => ['Metadata:size'],
  FILTER => "SingleColumnValueFilter('Metadata', 'size', =, 'binary:5000')"
}
```

```
hbase:1234:0> scan 'WebTable', {
hbase:1235:1*   COLUMNS => ['Metadata:size'],
hbase:1236:1*   FILTER => "SingleColumnValueFilter('Metadata', 'size', =, 'binary:5000')"
}
ROW
1-io.hekal:/page13
1-io.hekal:/page6
145-net.demo:/page14
145-net.demo:/page18
208-com.example:/page15
208-com.example:/page7
208-com.example:/page9
67-edu.sample:/page19
91-org.test:/page10
91-org.test:/page11
91-org.test:/page5
11 row(s)
Took 0.0301 seconds
hbase:1238:0>
```

- Find pages with HTTP error status codes

```
scan 'WebTable', {
  COLUMNS => ['Metadata:status'],
  FILTER => "SingleColumnValueFilter('Metadata', 'status', >=, 'binary:400')"
}
```

```
hbase:1238:0> scan 'WebTable', {
hbase:1239:1*   COLUMNS => ['Metadata:status'],
hbase:1240:1*   FILTER => "SingleColumnValueFilter('Metadata', 'status', >=, 'binary:400')"
}
ROW
1-io.hekal:/page1
1-io.hekal:/page13
1-io.hekal:/page6
145-net.demo:/page14
145-net.demo:/page18
208-com.example:/page20
67-edu.sample:/page16
91-org.test:/page11
8 row(s)
Took 0.0334 seconds
hbase:1242:0>
```

- List pages with outdated content (not modified in last 30 days)

```
scan 'WebTable', {
  COLUMNS => ['Metadata:modified'],
```

```
    FILTER => "SingleColumnValueFilter('Metadata', 'modified', <, 'binary:20250422')"
```

```
hbase:1246:0> scan 'WebTable', {
hbase:1247:1*   COLUMNS => ['Metadata:modified'],
hbase:1248:1*   FILTER => "SingleColumnValueFilter('Metadata', 'modified', <, 'binary:20250422')"
```

ROW	COLUMN+CELL
1-io.hekal:/page1	column=Metadata:modified, timestamp=2025-05-22T16:19:49.787, value=20240319
1-io.hekal:/page13	column=Metadata:modified, timestamp=2025-05-22T15:52:36.782, value=2025-05-12T15:24:41.159124
1-io.hekal:/page17	column=Metadata:modified, timestamp=2025-05-22T15:52:38.502, value=2025-02-22T03:43:30.253852
1-io.hekal:/page2	column=Metadata:modified, timestamp=2025-05-22T15:52:32.071, value=2025-03-15T07:59:11.086412
1-io.hekal:/page3	column=Metadata:modified, timestamp=2025-05-22T15:52:32.234, value=2025-02-07T13:26:16.361216
1-io.hekal:/page4	column=Metadata:modified, timestamp=2025-05-22T15:52:32.468, value=2025-04-08T23:11:52.165811
1-io.hekal:/page6	column=Metadata:modified, timestamp=2025-05-22T15:52:33.580, value=2025-05-16T16:41:06.504944
145-net.demo:/page14	column=Metadata:modified, timestamp=2025-05-22T15:52:37.371, value=2025-05-07T00:45:32.018432
145-net.demo:/page18	column=Metadata:modified, timestamp=2025-05-22T15:52:39.060, value=2025-01-26T20:38:43.598209
208-com.example:/page12	column=Metadata:modified, timestamp=2025-05-22T15:52:36.245, value=2025-04-09T02:22:23.697852
208-com.example:/page15	column=Metadata:modified, timestamp=2025-05-22T15:52:37.945, value=2025-04-07T07:36:05.500164
208-com.example:/page7	column=Metadata:modified, timestamp=2025-05-22T15:52:34.145, value=2025-02-26T16:37:51.904555
208-com.example:/page8	column=Metadata:modified, timestamp=2025-05-22T15:52:34.419, value=2025-04-01T03:11:45.488937
208-com.example:/page9	column=Metadata:modified, timestamp=2025-05-22T15:52:34.971, value=2025-01-07T13:30:06.172040
67-edu.sample:/page16	column=Metadata:modified, timestamp=2025-05-22T15:52:38.222, value=2025-04-12T08:07:21.043945
67-edu.sample:/page19	column=Metadata:modified, timestamp=2025-05-22T15:52:39.623, value=2025-01-27T06:15:08.624041
91-org.test:/page10	column=Metadata:modified, timestamp=2025-05-22T15:52:35.559, value=2025-05-04T17:44:53.454170
91-org.test:/page11	column=Metadata:modified, timestamp=2025-05-22T15:52:36.138, value=2025-01-28T21:58:52.196237
91-org.test:/page5	column=Metadata:modified, timestamp=2025-05-22T15:52:33.033, value=2025-02-04T06:54:46.200781

```
19 row(s)
Took 0.0681 seconds
hbase:1250:0>
```